

GUMBEL FLUCTUATIONS FOR THE COVER TIME OF NEAR-COLLISIONS IN DYSON BROWNIAN MOTION

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ABSTRACT. Let $\lambda(t)$ be a stationary Dyson Brownian motion, circular or of Ornstein–Uhlenbeck type, at inverse temperature $\beta > 1$, let τ_i be the first time the i -th spacing falls below ε , and let $\tau_{\text{cov}} = \max_i \tau_i$ be the cover time. The hazard of the i -th spacing is $\Lambda_i = \frac{2(\beta-1)}{\beta n} a_i \varepsilon^{\beta-1}$, where a_i is its collision coefficient, $\mathbb{P}(g_i \leq x) \sim a_i x^{\beta+1}/(\beta+1)$; for the circular ensemble a_i is a ratio of Morris integrals. We prove that when $\Lambda_i(\log n)^{2\beta+4} \rightarrow 0$ the normalized hitting times converge to a Poisson process with intensity $e^{-x} dx$, so that τ_{cov} has Gumbel fluctuations and each τ_i is asymptotically exponential. For the circular unitary ensemble this is unconditional: $\mathbb{P}(\frac{(n^2-1)\varepsilon}{24\pi} \tau_{\text{cov}} - \log n \leq x) \rightarrow e^{-e^{-x}}$ whenever $\varepsilon n^2 (\log n)^8 \rightarrow 0$. For other β and for the Hermite ensemble we assume a first-moment bound on the inverse square distances seen from a collision. Where the density varies over the index window, the centering acquires a $\log \log n$ correction. In the intermediate regime, where $\varepsilon n^2 \rightarrow \infty$ but $n\varepsilon \rightarrow 0$, a Poincaré inequality on mesoscopic windows with frozen exterior reduces the Gumbel law to a static hypothesis: the conditional hazard of a spacing, averaged under a local equilibrium measure with rigid boundary condition, equals Λ_i to relative precision $o(1/\log n)$. The method is spectral: survival of a set of spacings is controlled by the principal Dirichlet eigenvalue of the union of their near-collision sets, computed to relative precision $o(1/\log n)$ from capacity bounds along orthogonal directions of configuration space.

1. INTRODUCTION

1.1. The question. In the normalization

$$(1) \quad d\lambda_k = \sqrt{\frac{2}{\beta n}} dB_k + \frac{1}{n} \sum_{j \neq k} \frac{dt}{\lambda_k - \lambda_j} - \frac{\lambda_k}{2} dt, \quad k = 1, \dots, n,$$

Dyson Brownian motion is stationary under the Hermite β -ensemble, whose empirical measure approaches the semicircle law $\varrho(x) = \frac{1}{2\pi} \sqrt{4 - x^2}$. The spacing $g_i = \lambda_{i+1} - \lambda_i$ is of order $1/(n\varrho)$, diffuses with coefficient $4/(\beta n)$, and is pushed away from zero by the drift $2/(ng_i)$: near zero, ng_i behaves on the time scale $1/n$ like a Bessel process of dimension $\beta + 1$, which for $\beta > 1$ never hits the origin.

Fix $\varepsilon > 0$, an index set I , and let

$$\tau_i = \inf\{t \geq 0 : g_i(t) \leq \varepsilon\}, \quad \tau_{\text{cov}} = \max_{i \in I} \tau_i.$$

The cover time τ_{cov} is the first time at which every spacing in I has come within ε of a collision. If near-collisions are rare and weakly dependent, the spacings behave like the coupons of a coupon collector, and τ_{cov} should have Gumbel fluctuations around $\log |I|$ divided by the rate. We prove this, identify the rate, and describe the extremal process of the late hitting times.

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1.2. The hazard. Let p_i be the equilibrium density of g_i . The *collision coefficient*

$$(2) \quad a_i := \lim_{x \downarrow 0} x^{-\beta} p_i(x)$$

exists and is finite and positive (Lemma 2.3). The *hazard* of the i -th spacing is

$$(3) \quad \Lambda_i := \frac{2(\beta-1)}{\beta n} a_i \varepsilon^{\beta-1},$$

the flux through the level ε of a Bessel process of dimension $\beta+1$, diffusion coefficient $4/(\beta n)$ and stationary density $a_i x^\beta$ near the origin. In the bulk of the Hermite ensemble $a_i \approx (\beta+1)\kappa_\beta(n\varrho(\gamma_i))^{\beta+1}$, with γ_i the classical location and κ_β the small-spacing constant of the Sine $_\beta$ process ($\kappa_2 = \pi^2/9$), so $\Lambda_i \asymp n^\beta \varepsilon^{\beta-1}$.

1.3. Main results. The circular Dyson process is

$$(4) \quad d\theta_k = \sqrt{\frac{2}{\beta n}} dB_k + \frac{1}{2n} \sum_{j \neq k} \cot \frac{\theta_k - \theta_j}{2} dt, \quad k = 1, \dots, n,$$

stationary under the circular β -ensemble. Its n cyclic spacings ($g_n = \theta_1 + 2\pi - \theta_n$) share one collision coefficient, a ratio of Morris integrals (Proposition 6.1):

$$(5) \quad a_\beta(n) = \frac{(n-1)\Gamma(1+\frac{\beta}{2})^n}{2\pi\Gamma(1+\frac{n\beta}{2})} \prod_{j=0}^{n-3} \frac{\Gamma(\frac{\beta}{2}j+2\beta+1)\Gamma(\frac{\beta}{2}(j+1)+1)}{\Gamma(\frac{\beta}{2}j+\beta+1)^2\Gamma(1+\frac{\beta}{2})}, \quad a_2(n) = \frac{n(n^2-1)}{24\pi}.$$

Theorem A (Circular unitary ensemble). *Start (4) with $\beta=2$ from the circular unitary ensemble, let $\varepsilon_n n^2(\log n)^8 \rightarrow 0$, and put $\lambda_n = (n^2-1)\varepsilon_n/(24\pi)$. Then, with $\tau_{\text{cov}} = \max_{1 \leq i \leq n} \tau_i$,*

- (i) $\mathbb{P}(\lambda_n \tau_{\text{cov}} - \log n \leq x) \rightarrow \exp(-e^{-x})$ for every $x \in \mathbb{R}$;
- (ii) $\sum_{i=1}^n \delta_{\lambda_n \tau_i - \log n}$ converges in distribution, on $(x_0, \infty]$ for every $x_0 \in \mathbb{R}$, to the Poisson process with intensity $e^{-x} dx$;
- (iii) $\sup_{t \geq 0} |\mathbb{P}(\lambda_n \tau_i > t) - e^{-t}| \rightarrow 0$.

For general β and for the Hermite ensemble we need one estimate that we prove only for the circular unitary ensemble: a first-moment bound on the inverse square distances from a collision point to the other particles, under the law of the configuration conditioned on a collision (Hypothesis 2.4).

Theorem B (Circular ensemble). *Let $\beta > 1$, start (4) from the circular β -ensemble, assume Hypothesis 2.4 with $\kappa = 0$, and let $\lambda_n = \frac{2(\beta-1)}{\beta n} a_\beta(n) \varepsilon_n^{\beta-1}$ satisfy $\lambda_n (\log n)^{2\beta+4} \rightarrow 0$. Then (i)–(iii) of Theorem A hold.*

By Remark 6.2, $\lambda_n \asymp n^\beta \varepsilon_n^{\beta-1}$, so the condition is $\varepsilon_n \ll n^{-\beta/(\beta-1)}$ up to a power of $\log n$.

When the hazards differ, the centering is implicit. For an index set I_n let $t_n(x)$ solve

$$(6) \quad \sum_{i \in I_n} e^{-\Lambda_i t} = e^{-x}, \quad x > -\log |I_n|,$$

and let $x_n = t_n^{-1}$, with $x_n(0) = -\log |I_n|$.

Theorem C (Hermite ensemble). *Let $\beta > 1$, start (1) in equilibrium, let $I_n \subset \{1, \dots, n-1\}$, $|I_n| \rightarrow \infty$, satisfy Hypothesis 2.4, and let $\Lambda_{\max} = \max_{i \in I_n} \Lambda_i$. Assume either $\kappa = 0$ and $\Lambda_{\max} (\log n)^{2\beta+4} \rightarrow 0$, or $\Lambda_{\max} \leq n^{-\delta}$ with $\delta \geq 4(\beta+1)\kappa$. Then $\sum_{i \in I_n} \delta_{x_n(\tau_i)}$ converges in distribution on $(x_0, \infty]$ to the Poisson process with intensity $e^{-x} dx$; in particular $\mathbb{P}(\tau_{\text{cov}} \leq t_n(x)) \rightarrow \exp(-e^{-x})$, and $\sup_t |\mathbb{P}(\tau_i > t) - e^{-\Lambda_i t}| \rightarrow 0$ uniformly in $i \in I_n$.*

For the bulk window $I_n = [n/4, 3n/4]$ of the Gaussian unitary ensemble the explicit centering is not $\log n$ (Corollary 7.2):

$$\mathbb{P}(\Lambda_* \tau_{\text{cov}} - \log n + \log \log n - \log c_* \leq x) \rightarrow \exp(-e^{-x}),$$

where $\Lambda_* = \frac{\pi^2}{3} \varrho(\gamma_*)^3 n^2 \varepsilon$ is the smallest hazard, attained at the two ends of the window, $\gamma_* \approx -0.8079$ is the first quartile of the semicircle law, and $c_* = (4 - \gamma_*^2)^{3/2} / (3\pi|\gamma_*|) \approx 0.8042$. The last spacings to be hit sit near the ends of the window, where the density is smallest; their number is of order $n/\log n$, not n .

In the regime $\varepsilon n^2 \rightarrow \infty$, $n\varepsilon \rightarrow 0$, near-collisions are rare but more frequent than the relaxation of the whole spectrum, and the global spectral gap is useless. A Poincaré inequality on mesoscopic windows with frozen exterior replaces it. What remains is a static statement, Hypothesis 8.3: for every rigid exterior, the conditional hazard of a spacing averaged over its window equals Λ_i to relative precision $o(1/\log n)$.

Theorem D (Intermediate regime). *Let $\beta > 1$; in the Hermite case let $I_n \subset [an, bn]$ with $0 < a < b < 1$. Assume (P1)–(P2) of Hypothesis 2.4 with $\kappa = 0$ and Hypothesis 8.3 with parameters K_n, M_n . If $n\varepsilon_n(\log n)^{\beta/(\beta-1)} \rightarrow 0$ and $M_n^2 K_n \Lambda_{\max}(\log n)^2/n \rightarrow 0$, then $\mathbb{P}(\tau_{\text{cov}} \leq t_n(x)) \rightarrow \exp(-e^{-x})$, every order statistic of the τ_i has the corresponding limit, and $\sup_t |\mathbb{P}(\tau_i > t) - e^{-\Lambda_i t}| \rightarrow 0$.*

If Hypothesis 8.3 holds with $K_n = n^{2\xi}$ for every $\xi > 0$ and M_n a power of $\log n$, Theorem D covers every ε with $n\varepsilon \leq n^{-\delta}$: all of the intermediate regime up to its upper end.

1.4. Scope. The condition $\Lambda_i \ll (\log n)^{-2\beta-4}$ says that near-collisions are rarer than the relaxation of the whole spectrum, whose rate, the spectral gap, is a constant. The regime lies far below the static extremes: at a fixed time the smallest bulk spacing is of order $n^{-(\beta+2)/(\beta+1)}$ [6, 17], much larger than $n^{-\beta/(\beta-1)}$, so at a typical time no spacing is below ε and the cover time is a dynamical extreme. Between $n^{-\beta/(\beta-1)}$ and n^{-1} lies the intermediate regime of Theorem D; there the infinite-volume dynamics has no spectral gap [26], and Poincaré inequalities on mesoscopic windows take its place.

1.5. Method and related work. Cover times of random walks have Gumbel fluctuations by Poisson approximation of the number of unvisited sites [1, 5, 14]. For interacting particles the difficulty is joint survival of several spacings. We avoid couplings with independent Bessel processes and work with the Dirichlet form of the n -particle dynamics.

- (a) Survival of the spacings in S up to time t is avoidance of $A_S = \bigcup_{i \in S} \{g_i \leq \varepsilon\}$. Its probability lies between $\pi(f^2)e^{-t\mathcal{E}(f)/\pi(f^2)}$ for test functions f vanishing on A_S , by Jensen's inequality for the spectral measure, and $e^{-\lambda_{A_S} t}$, with λ_{A_S} the principal Dirichlet eigenvalue (Lemma 3.1). Matching the two to relative precision $o(1/\log n)$ gives the Gumbel law through factorial moments (Theorem 3.4). With a spectral gap, the killed semigroup has a single eigenvalue below the gap, as in Aldous–Brown [2], and multi-time survival factorizes; this gives the point process (Theorem 3.5). The gap of Dyson Brownian motion follows from Brascamp–Lieb (Proposition 2.1).
- (b) The gradients of non-adjacent spacings are orthogonal, so the energy of a function vanishing on A_S dominates the sum of its energies along these directions. Adjacent spacings meet at angle $2\pi/3$; the Gram matrix of a run of adjacent directions has norm below 2, which removes clusters (Proposition 4.2).
- (c) Along the direction of one spacing the Gibbs density is $g^\beta \Phi(g, y)$ with Φ decreasing in g (Lemma 2.2). This gives the exact upper bound $\mathbb{P}(g_i \leq x) \leq a_i x^{\beta+1}/(\beta+1)$ and reduces the lower bound to a moment of the inverse square distances seen from a collision, computed at $\beta = 2$ from the Palm kernel of a determinantal process.

- (d) In the intermediate regime the global gap is replaced by Brascamp–Lieb on windows of K particles with rigid exterior, whose Poincaré constant is of order n/K , and the eigenvalue bound by a decay estimate for $\|P_s^D \mathbf{1}_D\|^2$, which only involves functions with values in $[0, 1]$ (Section 8).

Hitting times of rare sets in reversible chains are approximately exponential [2, 3]; capacities and principal Dirichlet eigenvalues of small sets are the tools of metastability [11]. Small spacings of random matrices at a fixed time are described in [6, 16, 17, 8]. For random walks on vertex-transitive graphs, Gumbel fluctuations of the cover time are characterized by a diameter condition [7]. Poisson point processes for extremes and factorial moments are classical [24, 21].

1.6. Organization. Section 2 sets up the Dyson processes and collision coefficients. Section 3 proves the abstract limit theorems, Section 4 the capacity estimates, and Section 5 Theorems A–C. Section 6 treats the circular ensemble, Section 7 variable density, Section 8 the intermediate regime and Theorem D, and Section 9 clusters and open problems.

2. DYSON PROCESSES AS REVERSIBLE DIFFUSIONS

2.1. The two models. *Hermite.* Let $W = \{\lambda \in \mathbb{R}^n : \lambda_1 < \dots < \lambda_n\}$ and

$$\mu(d\lambda) = Z^{-1} e^{-V(\lambda)} d\lambda, \quad V(\lambda) = \frac{\beta n}{4} \sum_k \lambda_k^2 - \beta \sum_{j < k} \log(\lambda_k - \lambda_j).$$

The generator of (1) is $Lf = \frac{1}{\beta n} (\Delta f - \nabla V \cdot \nabla f)$, with Dirichlet form

$$(7) \quad \mathcal{E}(f) = \frac{1}{\beta n} \int |\nabla f|^2 d\mu.$$

For $\beta \geq 1$, (1) has a unique strong solution which never leaves W [4, Prop. 4.3.5]; it is the μ -symmetric diffusion associated with the closure $(\mathcal{E}, \mathcal{F})$ of (7) on $C_c^\infty(W)$, and since ∂W is polar, $\mathcal{F}$ contains the bounded Lipschitz functions on W [20].

Circular. Let $\theta_1 < \dots < \theta_n < \theta_1 + 2\pi$, $g_k = \theta_{k+1} - \theta_k$ for $k < n$ and $g_n = \theta_1 + 2\pi - \theta_n$. The spacings depend only on the projection of θ to $H = \{\sum_k \theta_k = 0\}$, and the projected process is a diffusion on the convex set $W^\circ = \{\theta \in H : \theta_1 < \dots < \theta_n < \theta_1 + 2\pi\}$, reversible for

$$\mu(d\theta) = Z^{-1} e^{-V(\theta)} d\theta, \quad V(\theta) = -\beta \sum_{j < k} \log \left| 2 \sin \frac{\theta_k - \theta_j}{2} \right|,$$

with Lebesgue measure on H and Dirichlet form (7), gradients taken in H [13]. The spacings have the law of the circular β -ensemble, invariant under cyclic relabelling.

2.2. Spectral gap.

Proposition 2.1. *For $\beta \geq 1$, $n \geq 2$ and $f \in \mathcal{F}$, $\text{Var}_\mu(f) \leq c_P^{-1} \mathcal{E}(f)$, with $c_P = \frac{1}{2}$ in the Hermite case and $c_P = \frac{1}{4}$ in the circular case.*

Proof. In both cases μ has density e^{-V} on an open convex set, with V smooth there and tending to $+\infty$ at the boundary. By the Brascamp–Lieb inequality [12], $\nabla^2 V \geq \rho$ implies $\text{Var}_\mu(f) \leq \rho^{-1} \int |\nabla f|^2 d\mu = \rho^{-1} \beta n \mathcal{E}(f)$. In the Hermite case $\nabla^2 V \geq \frac{\beta n}{2}$, since $-\log(\lambda_k - \lambda_j)$ is convex. In the circular case, for $v \in H$,

$$v^\top \nabla^2 V v = \frac{\beta}{4} \sum_{j < k} \frac{(v_k - v_j)^2}{\sin^2 \frac{\theta_k - \theta_j}{2}} \geq \frac{\beta}{4} \sum_{j < k} (v_k - v_j)^2 = \frac{\beta n}{4} |v|^2. \quad \square$$

2.3. Spacing coordinates and collision coefficients. Fix a spacing i . Let $w_i = \frac{1}{2}(e_{i+1} - e_i)$ (with $e_{n+1} = e_1$ in the circular case), so $\partial_{w_i} g_i = 1$ and $|w_i|^2 = \frac{1}{2}$. Write a configuration as (g, y) with $g = g_i$: in the Hermite case $y = (\lambda_1, \dots, \lambda_{i-1}, m, \lambda_{i+2}, \dots, \lambda_n)$, $m = \frac{1}{2}(\lambda_i + \lambda_{i+1})$, so $\lambda_i = m - \frac{g}{2}$, $\lambda_{i+1} = m + \frac{g}{2}$, with Jacobian one; in the circular case the same in an affine chart of H , the constant Jacobian absorbed in Z . Then $\partial_g = \partial_{w_i}$ and

$$(8) \quad e^{-V} = g^\beta \Phi_i(g, y),$$

where, with j ranging over the indices other than $i, i+1$ and up to factors free of g ,

$$\begin{aligned} \Phi_i(g, y) &= e^{-\frac{\beta n}{8} g^2} \prod_j \left| (m - \lambda_j)^2 - \frac{g^2}{4} \right|^\beta, \\ \Phi_i(g, y) &= \left(\frac{\sin(g/2)}{g/2} \right)^\beta \prod_j \left| 4 \sin^2 \frac{m - \theta_j}{2} - 4 \sin^2 \frac{g}{4} \right|^\beta \end{aligned}$$

in the Hermite and circular cases. The admissible g are $0 < g < g_{\max}(y)$, where $g_{\max}(y)/2$ is the distance from m to the nearest other particle. Let

$$D_i(y) = \sum_j (m - \lambda_j)^{-2} \quad \text{or} \quad D_i(y) = \sum_j \left(2 \sin \frac{m - \theta_j}{2} \right)^{-2}.$$

Lemma 2.2. *For admissible (g, y) , $\Phi_i(g, y) \leq \Phi_i(0, y)$. If moreover $g^2 D_i(y) \leq 2$, then $\Phi_i(g, y) \geq \Phi_i(0, y) \exp(-\frac{\beta g^2}{2}(D_i(y) + n))$.*

Proof. In the Hermite case $\Phi_i(g, y)/\Phi_i(0, y) = e^{-\beta n g^2/8} \prod_j (1 - u_j)^\beta$ with $u_j = g^2/(4(m - \lambda_j)^2) \in [0, 1)$, which is at most one; if $g^2 D_i \leq 2$ then $u_j \leq \frac{1}{2}$ and $\log(1 - u) \geq -2u$ gives the lower bound. In the circular case the ratio is $(\frac{\sin(g/2)}{g/2})^\beta \prod_j (1 - u_j)^\beta$ with $u_j = \sin^2 \frac{g}{4} / \sin^2 \frac{m - \theta_j}{2} \in [0, 1)$; now $2 \sum_j u_j \leq \frac{g^2}{2} D_i$, and $\frac{\sin x}{x} \geq e^{-x^2/3}$ for $0 \leq x \leq 1$. $\square$

Lemma 2.3. *Let $c_i(y) = Z^{-1} \Phi_i(0, y)$ for y with m strictly between its neighbours, and $a_i = \int c_i dy$. Then $0 < a_i < \infty$, $p_i(x) \leq a_i x^\beta$ for all $x > 0$, and $x^{-\beta} p_i(x) \rightarrow a_i$ as $x \downarrow 0$.*

Proof. Up to a constant, c_i is the density of an $(n-1)$ -particle Gibbs measure with a charge 2β at m , hence integrable. By (8), $p_i(x) = x^\beta \int Z^{-1} \Phi_i(x, y) \mathbf{1}\{x < g_{\max}(y)\} dy$; the integrand is at most $c_i(y)$ and tends to it as $x \downarrow 0$. $\square$

The collision law $\hat{\nu}_i(dy) = a_i^{-1} c_i(y) dy$ is the law of the other particles given that λ_i and λ_{i+1} have merged.

Hypothesis 2.4. There are $C_0 \geq 1$ and $\kappa \geq 0$ such that for all large n and all $i \in I_n$:

- (P1) $\max_{I_n} a_i \leq C_0 \min_{I_n} a_i$;
- (P2) $C_0^{-1} n^{\beta+1-\kappa} \leq a_i \leq C_0 n^{\beta+1+\kappa}$;
- (P3) $\int D_i d\hat{\nu}_i \leq C_0 n^{2+\kappa}$.

For the circular ensemble (P1) is an equality and (P2) holds with $\kappa = 0$ by (5); (P3) holds with $\kappa = 0$ for the circular unitary ensemble (Proposition 6.3).

3. RARE HITTING TIMES IN REVERSIBLE DIFFUSIONS

Let X be a diffusion on E , symmetric with respect to a probability measure π , with regular Dirichlet form $(\mathcal{E}, \mathcal{F})$, started from π . For a closed set A let $T_A = \inf\{t \geq 0 : X_t \in A\}$, $D = E \setminus A$, $\mathcal{F}_A = \{f \in \mathcal{F} : \hat{f} = 0 \text{ q.e. on } A\}$, and

$$\lambda_A = \inf\{\mathcal{E}(f)/\pi(f^2) : f \in \mathcal{F}_A, f \neq 0\}.$$

The part of X on D is associated with $(\mathcal{E}, \mathcal{F}_A)$, and $\mathbb{P}_\pi(T_A > t) = \langle \mathbf{1}_D, P_t^D \mathbf{1}_D \rangle_\pi$ with P^D the killed semigroup [20, Thms. 4.4.2–4.4.3].

Lemma 3.1. *For every closed A , every $t \geq 0$ and every $f \in \mathcal{F}_A$ with $0 \leq f \leq 1$,*

$$\pi(f^2) \exp\left(-t \frac{\mathcal{E}(f)}{\pi(f^2)}\right) \leq \mathbb{P}_\pi(T_A > t) \leq e^{-\lambda_A t}.$$

Moreover $\mathbb{E}_\pi T_A \leq 1/\lambda_A$ and $\mathbb{E}_\pi T_A \geq \pi(f)^2/\mathcal{E}(f)$.

Proof. Let ν be the spectral measure of $\mathbf{1}_D$ for the killed generator; it is carried by $[\lambda_A, \infty)$ and has mass $\pi(D) \leq 1$, so $\mathbb{P}_\pi(T_A > t) = \int e^{-st} \nu(ds) \leq e^{-\lambda_A t}$ and $\mathbb{E}_\pi T_A \leq 1/\lambda_A$. Since P_t^D preserves positivity and $0 \leq f \leq \mathbf{1}_D$, $\langle \mathbf{1}_D, P_t^D \mathbf{1}_D \rangle \geq \langle f, P_t^D f \rangle = \int e^{-st} \nu_f(ds)$, where ν_f has mass $\pi(f^2)$ and first moment $\mathcal{E}(f)$; Jensen's inequality gives the lower bound. Finally $u = \int_0^\infty P_t^D \mathbf{1}_D dt$ satisfies $\mathcal{E}(u) = \langle u, \mathbf{1}_D \rangle = \mathbb{E}_\pi T_A$ and $\pi(f) = \mathcal{E}(f, u) \leq \mathcal{E}(f)^{1/2} \mathcal{E}(u)^{1/2}$. $\square$

Lemma 3.2. *Assume $\text{Var}_\pi(f) \leq c_P^{-1} \mathcal{E}(f)$ on $\mathcal{F}$, and let $0 < \lambda_A < c_P$. Then λ_A is a simple isolated eigenvalue of the killed generator with eigenfunction $\varphi_A \geq 0$, $\|\varphi_A\| = 1$, the rest of the spectrum lies in $[c_P, \infty)$, and with $c_A = \langle \mathbf{1}_D, \varphi_A \rangle^2$,*

$$0 \leq \mathbb{P}_\pi(T_A > t) - c_A e^{-\lambda_A t} \leq (1 - c_A) e^{-c_P t}, \quad c_A \geq \lambda_A (\mathbb{E}_\pi T_A - c_P^{-1}).$$

Proof. Every two-dimensional subspace of $\mathcal{F}_A$ contains $f \neq 0$ with $\pi(f) = 0$, for which $\mathcal{E}(f) \geq c_P \pi(f^2)$; by min-max the spectral projection on $[0, c_P)$ has rank at most one. If φ is an eigenfunction, $|\varphi| \in \mathcal{F}_A$ has the same Rayleigh quotient, so $\varphi = \pm |\varphi|$. With ν as above, $\nu(\{\lambda_A\}) = c_A$ and $\nu([c_P, \infty)) \leq 1 - c_A$; this gives the first bound, and integrating in t gives $\mathbb{E}_\pi T_A \leq c_A/\lambda_A + c_P^{-1}$. $\square$

Lemma 3.3. *In the setting of Lemma 3.2, let $A_1 \supseteq \dots \supseteq A_k$ be closed with $0 < \lambda_{A_l} < c_P$, $0 = t_0 < t_1 < \dots < t_k$, $\Delta_l = t_l - t_{l-1}$. Then*

$$\mathbb{P}_\pi(X_s \notin A_l \text{ for } s \leq t_l, l \leq k) = \Theta \prod_l e^{-\lambda_{A_l} \Delta_l} + \rho,$$

with $1 - 5 \sum_l (1 - c_{A_l}) \leq \Theta \leq 1$ and $|\rho| \leq 2^k e^{-c_P \min_l \Delta_l}$.

Proof. Let $Q_l f = P_{\Delta_l}^{D_l}(f \mathbf{1}_{D_l})$. By the Markov property the probability is $\langle 1, Q_1 \dots Q_k 1 \rangle$. By Lemma 3.2, $Q_l = e^{-\lambda_{A_l} \Delta_l} \varphi_l \langle \varphi_l, \cdot \rangle + R_l$ with $\|R_l\| \leq e^{-c_P \Delta_l}$. The term without R_l 's is $\prod_l e^{-\lambda_{A_l} \Delta_l} \Theta$, $\Theta = \langle 1, \varphi_1 \rangle \langle \varphi_1, \varphi_2 \rangle \dots \langle \varphi_k, 1 \rangle$, and each of the other $2^k - 1$ terms is at most $e^{-c_P \min \Delta_l}$. All inner products lie in $[0, 1]$; $\|\varphi_l - 1\|^2 \leq 2(1 - c_{A_l})$ gives $\langle \varphi_l, \varphi_{l+1} \rangle \geq 1 - 2(1 - c_{A_l}) - 2(1 - c_{A_{l+1}})$, and $\langle 1, \varphi_l \rangle \geq c_{A_l}$. $\square$

Consider now a sequence of such diffusions with finite index sets I_n , $N_n = |I_n| \rightarrow \infty$, a graph on I_n of degree at most d , closed sets A_i and rates $\Lambda_i > 0$. For $S \subset I_n$ let $A_S = \bigcup_{i \in S} A_i$, $\lambda_S = \lambda_{A_S}$, $\Lambda_S = \sum_{i \in S} \Lambda_i$; S is *separated* if it contains no two neighbours. Let $\tau_i = T_{A_i}$, $\Lambda_{\max} = \max \Lambda_i$, $\Lambda_{\min} = \min \Lambda_i$, and t_n, x_n as in (6). Consider the conditions:

(A) $\Lambda_{\max} \leq C_0 \Lambda_{\min}$;

(B) for every m and D there are $\eta_n = \eta_n(m)$ with $\eta_n \log N_n \rightarrow 0$ and C_D such that for all S with $|S| \leq m$ and all $t \geq 0$,

$$(9) \quad \mathbb{P}_\pi(T_{A_S} > t) \leq \exp\left(-(1 - \eta_n)t \sum_C w_C\right) + C_D N_n^{-D}, \quad w_C = \left(1 + \frac{1}{3} \mathbf{1}_{\{|C| \geq 2\}}\right) \min_{i \in C} \Lambda_i,$$

the sum over the connected components C of S . By Lemma 3.1, (B) holds with $C_D = 0$ if

$$(10) \quad \lambda_S \geq (1 - \eta_n) \sum_C w_C;$$

(C) for every separated S with $|S| \leq m$ there is $f_S \in \mathcal{F}_{A_S}$, $0 \leq f_S \leq 1$, with $\pi(f_S^2) \geq 1 - \eta_n$ and $\mathcal{E}(f_S) \leq (1 + \eta_n) \Lambda_S \pi(f_S^2)$.

Theorem 3.4. *Under (A)–(C), for every $x \in \mathbb{R}$ the number of i with $\tau_i > t_n(x)$ converges in law to a Poisson variable with mean e^{-x} . Hence $\mathbb{P}(\max_i \tau_i \leq t_n(x)) \rightarrow \exp(-e^{-x})$, the k -th largest $\tau_{(k)}$ satisfies $\mathbb{P}(\tau_{(k)} \leq t_n(x)) \rightarrow e^{-e^{-x}} \sum_{j < k} e^{-jx}/j!$, and $\sup_t |\mathbb{P}(\tau_i > t) - e^{-\Lambda_i t}| \leq 3\eta_n(1)$ for all i .*

Proof. By Lemma 3.1, (B) and (C), for separated S with $|S| \leq m$,

$$(11) \quad (1 - \eta_n)e^{-(1+\eta_n)\Lambda_S t} \leq \mathbb{P}(T_{A_S} > t) \leq e^{-(1-\eta_n)\Lambda_S t} + C_D N_n^{-D}.$$

For $S = \{i\}$ this gives the last claim, the additive term being absorbed for large n . Since $N_n e^{-\Lambda_{\max} t} \leq \sum_i e^{-\Lambda_i t} \leq N_n e^{-\Lambda_{\min} t}$,

$$(12) \quad (\log N_n + x)/\Lambda_{\max} \leq t_n(x) \leq (\log N_n + x)/\Lambda_{\min}.$$

Fix x , $t = t_n(x)$, $W = \#\{i : \tau_i > t\}$ and $m \geq 1$. Then $\mathbb{E}(W)_m = \sum \mathbb{P}(T_{A_S} > t)$ over ordered m -tuples of distinct indices, S their set. There are at most N_n^m tuples, so with $D = m + 1$ the additive terms in (9) contribute $o(1)$ and may be dropped. For separated S , (11) and $\Lambda_S t \leq mC_0(\log N_n + x)$ give $\mathbb{P}(T_{A_S} > t) = (1 + o(1)) \prod_{i \in S} e^{-\Lambda_i t}$ uniformly. Summed over all tuples, with repetitions allowed, the product equals $(\sum_i e^{-\Lambda_i t})^m = e^{-mx}$; tuples with a repeated index or two neighbours contribute to this sum at most $m^2(d+1)e^{-\Lambda_{\min} t} e^{-(m-1)x} \rightarrow 0$, since $\Lambda_{\min} t \geq (\log N_n + x)/C_0$.

For non-separated S , (9) gives $\mathbb{P}(T_{A_S} > t) \leq \prod_C e^{-(1-\eta_n)w_C t}$ up to the dropped term. Group the tuples by the shape of their decomposition into components; the sum over placements factorizes over components. A singleton contributes $\sum_i e^{-(1-\eta_n)\Lambda_i t} \leq e^{\eta_n \Lambda_{\max} t} e^{-x} = O(1)$. A component of size $s \geq 2$ is a connected set containing its minimizing index, and there are at most $(ed)^s$ such sets per vertex, so it contributes at most $(ed)^s e^{-x} e^{\eta_n \Lambda_{\max} t} \max_i e^{-(\frac{1}{3} - \frac{4}{3}\eta_n)\Lambda_i t} \rightarrow 0$. Hence $\mathbb{E}(W)_m \rightarrow e^{-mx}$ for every m , and W converges to $\text{Poisson}(e^{-x})$. The statements on order statistics follow from $\{\tau_{(k)} \leq t\} = \{W < k\}$. $\square$

Theorem 3.5. *Assume (A), (C), the eigenvalue bound (10), $\text{Var}_\pi(f) \leq c_P^{-1} \mathcal{E}(f)$ with $c_P \geq c > 0$ for all n , and $\Lambda_{\max} \log N_n \rightarrow 0$. Then for every $x_0 \in \mathbb{R}$, $\sum_i \delta_{x_n(\tau_i)}$ converges in distribution on $(x_0, \infty]$ to the Poisson process with intensity $e^{-x} dx$.*

Proof. For separated S , $|S| \leq m$: by (C) and Lemma 3.1, $\lambda_S \leq (1 + \eta_n)\Lambda_S$ and, since $\pi(f_S) \geq \pi(f_S^2)$, $\mathbb{E}_\pi T_{A_S} \geq (1 - \eta_n)/((1 + \eta_n)\Lambda_S)$; with (10), $\lambda_S < c$ for large n and Lemma 3.2 gives $c_{A_S} \geq 1 - \eta'_n$, $\eta'_n \rightarrow 0$.

Fix $x_0 < a_1 < b_1 \leq a_2 < \dots < b_k \leq \infty$, $J_l = (t_n(a_l), t_n(b_l)]$, $M_l = \#\{i : \tau_i \in J_l\}$. We show $\mathbb{E} \prod_l (M_l)_{m_l} \rightarrow \prod_l (e^{-a_l} - e^{-b_l})^{m_l}$; by the method of moments the M_l then converge jointly to independent Poisson variables, which characterizes convergence to the Poisson process [21, Ch. 4]. The expectation is a sum over families of distinct indices of $\mathbb{P}(\tau_i \in J_{l(i)}, i \in S)$. Since $t'_n \geq 1/\Lambda_{\max}$, distinct endpoints satisfy $|t_n(a) - t_n(b)| \geq |a - b|/\Lambda_{\max}$, and $t_n(a_1) \geq (\log N_n + a_1)/\Lambda_{\max}$.

For separated S , expand $\mathbf{1}\{\tau \in J_l\} = \mathbf{1}\{\tau > t_n(a_l)\} - \mathbf{1}\{\tau > t_n(b_l)\}$. Each resulting term $\mathbb{P}(\tau_i > u_i, i \in S)$ is the probability of avoiding a decreasing sequence of sets A_{S_p} on $[0, s_p]$, $s_1 < s_2 < \dots$ the distinct values of the u_i . By Lemma 3.3 it equals $(1 + o(1)) \exp(-\sum_i \Lambda_i u_i) + O(2^m e^{-ch/\Lambda_{\max}})$ for some $h > 0$, because $\lambda_{S_p} = (1 + O(\eta_n))\Lambda_{S_p}$ and $\sum_i \Lambda_i u_i = O(\log N_n)$. Summing back, the main term is $\prod_i (e^{-\Lambda_i t_n(a_{l(i)})} - e^{-\Lambda_i t_n(b_{l(i)})})$, whose sum over all families with repetitions is $\prod_l (e^{-a_l} - e^{-b_l})^{m_l}$ by (6). Non-distinct or non-separated families, the $o(1)$ terms, and the terms $N_n^m 2^m e^{-ch/\Lambda_{\max}}$ (which vanish because $\Lambda_{\max} \log N_n \rightarrow 0$) are negligible exactly as in the proof of Theorem 3.4, with $t = t_n(a_1)$. $\square$

4. CAPACITY OF SPACING EVENTS

Return to the Dyson processes, with $A_i = \{g_i \leq \varepsilon\}$, i and $i \pm 1$ neighbours (indices mod n in the circular case), and hazards (3). Constants C depend only on β , C_0 and m .

Proposition 4.1. *Let S be separated, $|S| \leq m$, $0 < \varepsilon < R$, and $f = \prod_{i \in S} (1 - h(g_i))$ with $h = 1$ on $[0, \varepsilon]$, $h(g) = \frac{g^{1-\beta} - R^{1-\beta}}{\varepsilon^{1-\beta} - R^{1-\beta}}$ on $[\varepsilon, R]$, $h = 0$ on $[R, \infty)$. Then $f \in \mathcal{F}_{A_S}$, $0 \leq f \leq 1$,*

$$\mathcal{E}(f) \leq \frac{\Lambda_S}{1 - (\varepsilon/R)^{\beta-1}}, \quad \pi(f^2) \geq 1 - \sum_{i \in S} \frac{a_i R^{\beta+1}}{\beta + 1}.$$

Proof. f is bounded, Lipschitz and vanishes on A_S . Since $\nabla g_i = 2w_i$ and $w_i \perp w_j$ for non-neighbours, $|\nabla f|^2 \leq 2 \sum_{i \in S} h'(g_i)^2$. By Lemma 2.3, $\int h'(g_i)^2 d\mu \leq a_i \int_{\varepsilon}^R h'(g)^2 g^{\beta} dg = a_i(\beta - 1)\varepsilon^{\beta-1}/(1 - (\varepsilon/R)^{\beta-1})$. Finally $f = 1$ unless some $g_i < R$, and $\mu(g_i < R) \leq a_i R^{\beta+1}/(\beta + 1)$. $\square$

For a run C of s consecutive indices let $\mu_C = 1 + \cos \frac{\pi}{s+1}$, and let $\mathcal{C}(S)$ be the decomposition of S into runs.

Proposition 4.2. *Assume Hypothesis 2.4. Let $K \geq 1$ and $R > 0$ with $8K(nR)^2 \leq 1$. For every $S \subset I_n$ with $|S| \leq m$,*

$$\lambda_S \geq (1 - \eta) \sum_{C \in \mathcal{C}(S)} \mu_C^{-1} \sum_{i \in C} \Lambda_i, \quad \eta = C \left(K(nR)^2 + \frac{n^{\kappa}}{K} + \left(\frac{\Lambda_{\max} n^{\kappa}}{(nR)^{\beta+1}} \right)^{1/2} + \Lambda_{\max} \right).$$

Proof. Geometry. Let $u_i = \sqrt{2}w_i$. For $|i - j| \geq 2$, $u_i \perp u_j$, and $\langle u_i, u_{i+1} \rangle = -\frac{1}{2}$; the Gram matrix of a run C is tridiagonal with largest eigenvalue μ_C , and different runs span orthogonal subspaces. Hence $|v|^2 \geq \sum_C \mu_C^{-1} \sum_{i \in C} \langle u_i, v \rangle^2$, and with $\langle u_i, \nabla f \rangle^2 = 2(\partial_{g_i} f)^2$,

$$(13) \quad \mathcal{E}(f) \geq \sum_C \mu_C^{-1} \sum_{i \in C} \mathcal{E}_i(f), \quad \mathcal{E}_i(f) = \frac{2}{\beta n} \int (\partial_{g_i} f)^2 d\mu.$$

One direction. Let $f \in \mathcal{F}_{A_S}$. If $\pi(f) = 0$, then $\mathcal{E}(f) \geq c_P \pi(f^2)$ and there is nothing to prove; if $\mathcal{E}(f) > 2\Lambda_S \pi(f)^2$, go to the conclusion. Otherwise fix $i \in S$ and $G = \{y : D_i(y) \leq Kn^2\}$. For $y \in G$ and $g \leq 2R$, $g^2 D_i \leq 2$, so (g, y) is admissible and, by Lemma 2.2, $Z^{-1} g^{\beta} \Phi_i(g, y) \geq e^{-\eta_1} c_i(y) g^{\beta}$ with $\eta_1 \leq 8\beta K(nR)^2$. For almost every y , $g \mapsto f(g, y)$ is absolutely continuous and vanishes at ε , so for $r \in [R, 2R]$

$$f(r, y)^2 \leq \int_{\varepsilon}^r (\partial_g f)^2 g^{\beta} dg \int_{\varepsilon}^r g^{-\beta} dg \leq \frac{\varepsilon^{1-\beta}}{\beta - 1} \int_{\varepsilon}^r (\partial_g f)^2 g^{\beta} dg.$$

Averaging over r with weight r^{β}/M_R , $M_R = \int_R^{2R} r^{\beta} dr \geq R^{\beta+1}$, and setting $\sigma(dr dy) = M_R^{-1} r^{\beta} c_i(y) \mathbf{1}_G(y) \mathbf{1}_{[R, 2R]}(r) dr dy$,

$$\mathcal{E}_i(f) \geq e^{-\eta_1} \frac{2(\beta-1)}{\beta n} \varepsilon^{\beta-1} \int f^2 d\sigma, \quad \sigma(1) \geq a_i(1 - \eta_P), \quad \eta_P = \hat{\nu}_i(D_i > Kn^2) \leq \frac{C_0 n^{\kappa}}{K}.$$

On the slab $\sigma \leq e^{\eta_1} M_R^{-1} \mu$, so by Proposition 2.1, $\|f - \pi(f)\|_{L^2(\sigma)}^2 \leq e^{\eta_1} \mathcal{E}(f)/(c_P M_R) \leq 2e^{\eta_1} \Lambda_S \pi(f)^2/(c_P M_R)$, and

$$\left(\int f^2 d\sigma \right)^{1/2} \geq |\pi(f)| \sqrt{a_i} (\sqrt{1 - \eta_P} - \sqrt{\eta_2}), \quad \eta_2 = \frac{2e^{\eta_1} \Lambda_S}{c_P a_i M_R} \leq \frac{C \Lambda_{\max} n^{\kappa}}{(nR)^{\beta+1}},$$

using $\Lambda_S/a_i \leq mC_0 \frac{2(\beta-1)}{\beta n} \varepsilon^{\beta-1} \leq mC_0^2 \Lambda_{\max} n^{\kappa-\beta-1}$ by (P1), (P2). Thus $\mathcal{E}_i(f) \geq (1 - \eta_1 - \eta_P - 2\sqrt{\eta_2}) \Lambda_i \pi(f)^2$.

Conclusion. By (13), $\mathcal{E}(f) \geq (1 - \eta') \Lambda_S^{\text{eff}} \pi(f)^2$ with $\Lambda_S^{\text{eff}} = \sum_C \mu_C^{-1} \sum_{i \in C} \Lambda_i \leq \Lambda_S$, in all cases. Since $\pi(f^2) \leq \pi(f)^2 + c_P^{-1} \mathcal{E}(f)$, $\mathcal{E}(f)/\pi(f^2) \geq (1 - \eta' - 4m\Lambda_{\max}) \Lambda_S^{\text{eff}}$. $\square$

Corollary 4.3. *Assume Hypothesis 2.4. Conditions (A), (C) and the eigenvalue bound (10) of Section 3 hold for the sets A_i , $i \in I_n$, with $\eta_n \log n \rightarrow 0$, in each of the cases:*

- (i) $\kappa = 0$ and $\Lambda_{\max}(\log n)^{2\beta+4} \rightarrow 0$;
- (ii) $\Lambda_{\max} \leq n^{-\delta}$ and $\kappa \leq \delta/(4(\beta+1))$.

Proof. (A) is (P1). For runs of length $s \geq 2$, $s/\mu_C \geq \frac{4}{3}$, so Proposition 4.2 gives (B). In case (i) take $L = \log n$, $K = L^2$, $nR = L^{-2}$: then $\eta \leq C(L^{-2} + (\Lambda_{\max} L^{2\beta+2})^{1/2} + \Lambda_{\max})$ and $\eta L \rightarrow 0$. In case (ii) take $K = n^\theta$, $nR = n^{-\theta}$, $\theta = \delta/(2(\beta+1))$: then $\eta \leq Cn^{-\theta/2}$. For (C) use Proposition 4.1 with the same R : $\sum_i a_i R^{\beta+1} \leq mC_0 n^\kappa (nR)^{\beta+1}$, and since $(n\varepsilon)^{\beta-1} \leq C\Lambda_{\max} n^{\kappa-1}$ by (P2), $(\varepsilon/R)^{\beta-1} = (n\varepsilon)^{\beta-1} (nR)^{1-\beta}$ is $O(n^{-1/2})$ in both cases. $\square$

5. PROOFS OF THEOREMS A–C

Proof of Theorems B and C. By Corollary 4.3, (A)–(C) hold with $\eta_n \log n \rightarrow 0$; by Proposition 2.1 the spectral gap is at least $\frac{1}{4}$; and $\Lambda_{\max} \log n \rightarrow 0$. Theorems 3.4 and 3.5 give Theorem C. In the circular case all hazards equal λ_n , so $x_n(\tau) = \lambda_n \tau - \log n$, which is Theorem B. $\square$

Proof of Theorem A. By Proposition 6.1, $a_2(n) = n(n^2-1)/(24\pi)$, so (P1)–(P2) hold with $\kappa = 0$, and (P3) holds with $\kappa = 0$ by Proposition 6.3. The hazard is $\lambda_n = (n^2-1)\varepsilon_n/(24\pi)$, and $\lambda_n(\log n)^8 \rightarrow 0$; apply Theorem B. $\square$

6. THE CIRCULAR ENSEMBLE

Proposition 6.1. *The collision coefficient of the circular β -ensemble is (5).*

Proof. Let $P = Z_n^{-1} \prod_{j < k} |e^{i\theta_j} - e^{i\theta_k}|^\beta$ on $[0, 2\pi)^n$, where $Z_n = (2\pi)^n \Gamma(1 + \frac{n\beta}{2}) / \Gamma(1 + \frac{\beta}{2})^n$ [19]. All factors of P are bounded, so by dominated convergence and rotation invariance

$$\lim_{x \downarrow 0} x^{-\beta} \rho_2(\phi, \phi + x) = \frac{n(n-1)(2\pi)^{n-2}}{Z_n} M_{n-2}(\beta, \beta, \frac{\beta}{2}),$$

where, by Morris' identity [19],

$$\begin{aligned} M_N(a, b, \lambda) &= \int_{[-\pi, \pi]^N} \prod_j e^{\frac{i}{2}(a-b)\theta_j} |1 + e^{i\theta_j}|^{a+b} \prod_{j < k} |e^{i\theta_j} - e^{i\theta_k}|^{2\lambda} \frac{d\theta}{(2\pi)^N} \\ &= \prod_{j=0}^{N-1} \frac{\Gamma(\lambda j + a + b + 1) \Gamma(\lambda(j+1) + 1)}{\Gamma(\lambda j + a + 1) \Gamma(\lambda j + b + 1) \Gamma(1 + \lambda)}. \end{aligned}$$

For small x , pairs at distance x are consecutive up to $O(x^{2\beta+1})$, and the n spacings are identically distributed, so $np(x) = 2\pi\rho_2(0, x) + O(x^{2\beta+1})$, which gives (5). At $\beta = 2$ the product telescopes, $\prod_{j=0}^{n-3} \frac{(j+4)(j+3)}{j+2} = \frac{n(n+1)!}{12}$, and $a_2(n) = n(n^2-1)/(24\pi)$. $\square$

Remark 6.2. By Stirling's formula, $(2\pi)^{\beta+1} n^{-\beta-1} a_\beta(n) \rightarrow C_\beta$, where

$$C_\beta = (\pi\beta)^\beta \frac{\Gamma(1 + \frac{\beta}{2})^3}{\Gamma(1 + \beta) \Gamma(1 + \frac{3\beta}{2})}$$

is the small-distance coefficient of the two-point function of Sine_β [18, Ch. 13]: $C_1 = \pi^2/6$, $C_2 = \pi^2/3$, $C_4 = 16\pi^4/135$, and $\kappa_\beta = C_\beta/(\beta+1)$. Hence $\lambda_n \sim \frac{2(\beta-1)}{\beta} C_\beta (2\pi)^{-\beta-1} n^\beta \varepsilon^{\beta-1}$.

As a law of $n-2$ unordered points, the collision law of the circular ensemble is the law Q_m with density proportional to $\prod_j |e^{im} - e^{i\theta_j}|^{2\beta} \prod_{j < k} |e^{i\theta_j} - e^{i\theta_k}|^\beta$, averaged over a uniform m .

Proposition 6.3. *For the circular unitary ensemble, $\int D_i d\hat{y}_i \leq n^2$.*

Proof. The CUE is determinantal with kernel $K(x, y) = \langle v_y, v_x \rangle$, $v_x = (2\pi)^{-1/2}(e^{-ikx})_{k < n} \in \mathbb{C}^n$, and $\rho_3(p, q, z)/\rho_2(p, q) = \text{dist}(v_z, \text{span}\{v_p, v_q\})^2$. Letting $p, q \rightarrow m$ as in Proposition 6.1, the one-point density of Q_m is $\rho^{(m)}(z) = \text{dist}(v_z, \text{span}\{v_m, \partial v_m\})^2$. Hence $\rho^{(m)}(z) \leq |v_z|^2 = \frac{n}{2\pi}$ and, for $u = z - m \in [-\pi, \pi]$, $\rho^{(m)}(z) \leq |v_z - v_m - u\partial v_m|^2 \leq \frac{u^4}{4} \sup |\partial^2 v|^2 \leq \frac{n^5 u^4}{40\pi}$. With $(2 \sin \frac{u}{2})^{-2} \leq \pi^2/(4u^2)$,

$$\int D dQ_m \leq \int_{|u| \leq 1/n} \frac{n^5 u^4}{40\pi} \cdot \frac{\pi^2}{4u^2} du + \int_{1/n \leq |u| \leq \pi} \frac{n}{2\pi} \cdot \frac{\pi^2}{4u^2} du \leq \frac{\pi n^2}{240} + \frac{\pi n^2}{4} \leq n^2. \quad \square$$

Remark 6.4. For the Gaussian unitary ensemble the same argument with the Hermite kernel bounds the inverse square distances seen from a collision at a bulk point by Cn^2 . Hypothesis 2.4 concerns the collision of a labelled spacing, so it also needs a collision of the i -th spacing to occur within $n^{-1+o(1)}$ of γ_i (rigidity of the conditioned ensemble) and $a_i \asymp n^{\beta+1}$; both are expected for every β .

7. VARIABLE DENSITY AND THE EXPLICIT CENTERING

Proposition 7.1. *Let $0 < a < b < 1$, $I_n = \{i : an \leq i \leq bn\}$, and $\Lambda_i = r_n \psi(i/n)(1 + o(1/\log n))$ uniformly, where $\psi \in C^2[a, b]$ is positive and attains its minimum ψ_* only on a nonempty $E_* \subseteq \{a, b\}$, with $\psi' \neq 0$ on E_* . With $c_\psi = \sum_{e \in E_*} |\psi'(e)|^{-1}$,*

$$\psi_* r_n t_n(x) = \log n - \log \log n + \log(\psi_* c_\psi) + x + o(1).$$

Proof. Put $s = r_n t_n(x) \asymp \log n$ by (12). The errors in Λ_i change $\Lambda_i t_n(x)$ by $o(1)$. Since $e^{-s\psi}$ varies on the scale $1/s \gg 1/n$, $\sum_{I_n} e^{-s\psi(i/n)} = (1 + o(1))n \int_a^b e^{-s\psi}$, and Laplace's method gives $\int_a^b e^{-s\psi} = e^{-s\psi_*} c_\psi s^{-1}(1 + O(s^{-1}))$. Thus $s\psi_* = \log n + \log c_\psi - \log s + x + o(1)$ and $\log s = \log \log n - \log \psi_* + o(1)$. $\square$

Corollary 7.2. *Let $\beta = 2$, $I_n = \{i : \frac{n}{4} \leq i \leq \frac{3n}{4}\}$, assume the hypotheses of Theorem C and $a_i = \frac{\pi^2}{3}(n\varrho(\gamma_i))^3(1 + o(1/\log n))$ uniformly. Then $\mathbb{P}(\Lambda_* \tau_{\text{cov}} - \log n + \log \log n - \log c_* \leq x) \rightarrow \exp(-e^{-x})$, with Λ_* , c_* as in the introduction.*

Proof. Here $\Lambda_i = r_n \psi(i/n)(1 + o(1/\log n))$ with $r_n = \frac{\pi^2}{3}n^2\varepsilon$ and $\psi(u) = \varrho(\gamma(u))^3$, $\gamma(u)$ the u -quantile of ϱ . Since $\gamma' = 1/\varrho(\gamma)$, $\psi' = 3\varrho\varrho'(\gamma)$; ψ is minimal exactly at $\frac{1}{4}$ and $\frac{3}{4}$, with $|\psi'| = 3\varrho(\gamma_*)|\varrho'(\gamma_*)|$ at both. So $\psi_* c_\psi = 2\varrho(\gamma_*)^2/(3|\varrho'(\gamma_*)|) = c_*$, and Theorem C with Proposition 7.1 concludes. $\square$

The correction is absent for constant hazards, as in the circular ensemble or on mesoscopic windows of $o(n/\log n)$ indices.

8. THE INTERMEDIATE REGIME

Theorem 3.4 needs no spectral gap. In Section 4 the gap enters only through Proposition 4.2, which bounds the energy of a function vanishing on A_S by comparing it with its mean. Here we replace the global gap by a Poincaré inequality on a mesoscopic window with frozen exterior, and the eigenvalue bound by a decay estimate that only involves functions with values in $[0, 1]$.

8.1. The conditional hazard.

Lemma 8.1. *For a spacing i and fixed y , let $\ell = g_{\max}(y)$, $w(g) = Z^{-1}g^\beta \Phi_i(g, y)$ on $(0, \ell)$, $M(g) = \int_g^\ell w$, and*

$$J_i(y) = \int_\varepsilon^\ell \frac{M(g)}{w(g)} dg, \quad H_i(y) = \frac{2}{\beta n J_i(y)}.$$

Then $\mathcal{E}_i(f) \geq \int H_i f^2 d\mu$ for every $f \in \mathcal{F}_{A_i}$.

Proof. On almost every line $u(g) = f(g, y)$ is absolutely continuous with $u(\varepsilon) = 0$, so $u(g)^2 \leq \int_\varepsilon^g u'^2 w \int_\varepsilon^g w^{-1}$. Multiplying by $w(g)$ and integrating, Fubini gives $\int_\varepsilon^\ell u^2 w \leq J_i(y) \int_\varepsilon^\ell u'^2 w$; integrate over y , using $f = 0$ for $g \leq \varepsilon$. $\square$

H_i is the hazard of the i -th spacing conditionally on all other coordinates. For typical y the integral $J_i(y)$ is dominated by g near ε , where $w(g) \approx c_i(y)g^\beta$ and $M(g) \approx m(y) = \int_0^\ell w$; hence $H_i(y) \approx \Lambda_i c_i(y)/(a_i m(y))$, a function with μ -mean close to Λ_i that fluctuates by a factor of order one with the positions of the neighbouring particles.

8.2. Windows. Fix $K \geq 1$ and, for $i \in I_n$, the window $W_i = \{i - K, \dots, i + K + 1\}$ of particles (indices mod n in the circular case). Write a configuration as (x, z) with x the window particles and z the others, let μ_z be the conditional law of x given z , and let

$$\mathcal{E}_z(f) = \frac{1}{\beta n} \int |\nabla_x f|^2 d\mu_z, \quad \int \mathcal{E}_z(f) \mu(dz) \leq \mathcal{E}(f).$$

For $\xi > 0$ the exterior z is *rigid* if $|(z_j - z_{j'}) - (\gamma_j - \gamma_{j'})| \leq n^{-1+\xi}$ for all $j, j' \notin W_i$ with $|j - i|, |j' - i| \leq 3K$, where γ_j are the classical locations ($\gamma_j = 2\pi j/n$ in the circular case). By rigidity of β -ensembles in the bulk [9, 10] and of the circular β -ensemble [22], for every $\xi, D > 0$ the probability that the exterior of W_i is not rigid is at most $C_{\xi, D} n^{-D}$, uniformly in i when $I_n \subset [an, bn]$, $0 < a < b < 1$, in the Hermite case.

Lemma 8.2. *Let $K \geq n^{2\xi}$ and $3K < n/2$. There is C depending on a, b such that for every rigid exterior z and every f , $\text{Var}_{\mu_z}(f) \leq \frac{CK}{n} \mathcal{E}_z(f)$.*

Proof. The measure μ_z has density proportional to e^{-V_z} on the convex set of ordered window configurations between $z_- = z_{i-K-1}$ and $z_+ = z_{i+K+2}$, with $V_z(x) = \sum_a U(x_a) - \beta \sum_{a < b} \log |x_a - x_b|$ (with $2 \sin \frac{x_a - x_b}{2}$ in the circular case) and U the confining part plus $-\beta \sum_{j \notin W_i} \log |x - z_j|$. The pair interaction is convex. For $x \in (z_-, z_+)$, rigidity gives $z_{i+K+1+k} - x \leq CK/n$ for $1 \leq k \leq K$, so

$$U''(x) \geq \beta \sum_{k=1}^K (z_{i+K+1+k} - x)^{-2} \geq \frac{\beta n^2}{C^2 K},$$

using $\frac{d^2}{du^2}(-\log |2 \sin \frac{u}{2}|) = \frac{1}{4 \sin^2(u/2)} \geq u^{-2}$ in the circular case. Thus $\nabla^2 V_z \geq \beta n^2/(C^2 K)$, and Brascamp–Lieb gives $\text{Var}_{\mu_z}(f) \leq \frac{C^2 K}{\beta n^2} \int |\nabla_x f|^2 d\mu_z = \frac{CK}{n} \mathcal{E}_z(f)$. $\square$

Hypothesis 8.3. There are $\xi > 0$, $K = K_n \geq n^{2\xi}$, $M = M_n \geq 1$ and η_n with $\eta_n \log n \rightarrow 0$ such that for all $i \in I_n$ and every rigid exterior z of W_i ,

$$\int \min(H_i, M\Lambda_i) d\mu_z \geq (1 - \eta_n)\Lambda_i.$$

Hypothesis 8.3 is static: it concerns the local equilibrium measures μ_z of Bourgade–Erdős–Yau [9] with good boundary conditions, and asks that the averaged conditional hazard be insensitive to the boundary condition. Since $\int H_i d\mu_z \approx \frac{2(\beta-1)}{\beta n} \varepsilon^{\beta-1} a_i(z)$, where $a_i(z)$ is the collision coefficient of the i -th spacing under μ_z , and $\int a_i(z) \mu(dz) = a_i$, the hypothesis says that $a_i(z) = a_i(1 + o(1/\log n))$ uniformly over rigid exteriors. Its proof amounts to a local law for the collision statistics of local equilibrium measures, uniform in the boundary condition; the uniqueness theorem for these measures [9] gives the corresponding statement for gap statistics at a fixed positive scale.

8.3. Decay of survival.

Lemma 8.4. *Let A be closed and suppose $\mathcal{E}(f) \geq a\pi(f^2) - b$ for every $f \in \mathcal{F}_A$ with $0 \leq f \leq 1$, where $a > 0$, $b \geq 0$. Then $\mathbb{P}_\pi(T_A > t) \leq e^{-at} + b/a$ for all $t \geq 0$.*

Proof. Let $u_s = P_s^D \mathbf{1}_D$ and $\phi(s) = \|u_s\|^2$. For $s > 0$, $u_s \in \mathcal{F}_A$, $0 \leq u_s \leq 1$, and $\phi'(s) = -2\mathcal{E}(u_s) \leq -2a\phi(s) + 2b$, so $\phi(s) \leq e^{-2as} + b/a$. Since P^D is self-adjoint, $\mathbb{P}_\pi(T_A > t) = \langle \mathbf{1}_D, P_t^D \mathbf{1}_D \rangle = \phi(t/2)$. $\square$

Proposition 8.5. *Assume (P1), Hypothesis 8.3, and rigidity as above. For every D there is C such that for every $S \subset I_n$ with $|S| \leq m$ and every $f \in \mathcal{F}_{A_S}$ with $0 \leq f \leq 1$,*

$$\mathcal{E}(f) \geq \left(1 - \eta_n - CM \left(\frac{K\Lambda_S}{n}\right)^{1/2} - CM \frac{K\Lambda_S}{n}\right) \Lambda_S^{\text{eff}} \pi(f^2) - C\Lambda_S n^{-D},$$

with $\Lambda_S^{\text{eff}} = \sum_C \mu_C^{-1} \sum_{i \in C} \Lambda_i$ as in Proposition 4.2.

Proof. If $\mathcal{E}(f) > 2\Lambda_S \pi(f^2)$ there is nothing to prove, so assume the contrary. By (13) and Lemma 8.1 it suffices to show, for $i \in S$,

$$\int H_i f^2 d\mu \geq \left(1 - \eta_n - CM \left(\frac{K\Lambda_S}{n}\right)^{1/2} - CM \frac{K\Lambda_S}{n}\right) \Lambda_i \pi(f^2) - C\Lambda_i n^{-D}.$$

Let $h = \min(H_i, M\Lambda_i) \leq H_i$ and write $\int h f^2 d\mu = \int \mu(dz) [\mu_z(h) \mu_z(f^2) + \text{Cov}_{\mu_z}(h, f^2)]$. Drop the non-negative contribution of non-rigid exteriors. For rigid z , $\mu_z(h) \geq (1 - \eta_n) \Lambda_i$ by Hypothesis 8.3, and, since $0 \leq h \leq M\Lambda_i$ and $0 \leq f \leq 1$,

$$|\text{Cov}_{\mu_z}(h, f^2)| \leq M\Lambda_i \|f^2 - \mu_z(f^2)\|_{L^1(\mu_z)} \leq M\Lambda_i (2\|f - \mu_z(f)\|_{L^2(\mu_z)} \|f\|_{L^2(\mu_z)} + \text{Var}_{\mu_z}(f)).$$

By Lemma 8.2, $\text{Var}_{\mu_z}(f) \leq CK\mathcal{E}_z(f)/n$. Integrating over z with Cauchy–Schwarz, $\int \|f - \mu_z(f)\|_{L^2(\mu_z)} \|f\|_{L^2(\mu_z)} \mu(dz) \leq (CK\mathcal{E}(f)/n)^{1/2} \pi(f^2)^{1/2}$, and $\int_{\text{rigid}} \mu_z(f^2) \mu(dz) \geq \pi(f^2) - C_D n^{-D}$ because $f^2 \leq 1$. With $\mathcal{E}(f) \leq 2\Lambda_S \pi(f^2)$ this is the claim. $\square$

Proof of Theorem D. Condition (A) of Section 3 is (P1). For (B), apply Lemma 8.4 to A_S with the inequality of Proposition 8.5: $a = (1 - \eta'_n) \Lambda_S^{\text{eff}}$, $\eta'_n = \eta_n + CM(K\Lambda_{\max}/n)^{1/2}$, and $b = C\Lambda_S n^{-D}$, so $b/a \leq Cn^{-D}$; then (9) follows from $\Lambda_S^{\text{eff}} \geq \sum_C w_C$, and $\eta'_n \log n \rightarrow 0$ because $M^2 K \Lambda_{\max} (\log n)^2 / n \rightarrow 0$. For (C), Proposition 4.1 with $nR = (\log n)^{-1}$ gives the test functions with $\eta_n \leq C((\log n)^{-\beta-1} + (n\varepsilon \log n)^{\beta-1})$, and $\eta_n \log n \rightarrow 0$ because $n\varepsilon (\log n)^{\beta/(\beta-1)} \rightarrow 0$. Theorem 3.4 concludes. $\square$

Since $\Lambda_i \asymp n(n\varepsilon)^{\beta-1}$, the condition $M^2 K \Lambda_{\max} (\log n)^2 / n \rightarrow 0$ of Theorem D reads

$$M^2 K (n\varepsilon)^{\beta-1} (\log n)^2 \rightarrow 0.$$

If Hypothesis 8.3 holds with $K = n^{2\xi}$ for every $\xi > 0$ and M a power of $\log n$, Theorem D covers every ε with $n\varepsilon \leq n^{-\delta}$, for any $\delta > 0$: all of the intermediate regime up to its upper end $n\varepsilon \rightarrow 0$. The infinite-volume Dyson Brownian motion has no spectral gap [26]; the window inequality of Lemma 8.2 replaces it, at the price of the static Hypothesis 8.3.

9. CLUSTERS AND OPEN PROBLEMS

9.1. Adjacent spacings. Clusters of adjacent spacings are removed in Theorem 3.4 by the factor $\frac{4}{3}$ of (10). Statically they are much rarer: two adjacent spacings are both below s with probability of order $s^{3\beta+2}$, not $s^{2\beta+2}$. At $\beta = 2$ the constant is explicit.

Proposition 9.1. *For the Sine₂ process with unit density, and g_1, g_2 the two spacings following the point at the origin under the Palm measure,*

$$\mathbb{P}(g_1 \leq s, g_2 \leq s) \sim \frac{31\pi^6}{16200} s^8, \quad s \downarrow 0.$$

Proof. Let $K(x, y) = \frac{\sin \pi(x-y)}{\pi(x-y)}$. The Palm measure at 0 is determinantal with kernel $K_0(x, y) = K(x, y) - K(x, 0)K(0, y)$ [25]. Its two-point function $\rho(a, b) = K_0(a, a)K_0(b, b) - K_0(a, b)^2$ vanishes to second order on $\{a = 0\}$, $\{b = 0\}$ and $\{a = b\}$, and expanding the sine kernel at the origin gives $\rho(a, b) = \frac{\pi^6}{135}a^2b^2(a-b)^2 + O(|(a, b)|^8)$. By inclusion-exclusion for determinantal processes, $\mathbb{P}(g_1 \leq s, g_2 \leq s)$ equals the integral of ρ over $\{0 < a < s, a < b < a + s\}$ up to terms of higher order, and $\int_0^s \int_0^s a^2 u^2 (a + u)^2 du da = \frac{31}{120}s^8$. $\square$

9.2. Open problems.

- (1) Hypothesis 8.3, which by Theorem D gives the Gumbel law throughout $n\varepsilon \leq n^{-\delta}$; the point process in that regime also needs multi-time survival bounds without a gap.
- (2) Hypothesis 2.4 for the Hermite ensemble and for $\beta \neq 2$ (Remark 6.4).
- (3) The recurrent case $\beta = 1$: the one-dimensional capacity becomes $\int_\varepsilon^R g^{-1} dg$, the hazard is of order $a_i/(n \log(1/(n\varepsilon)))$, and the cut-off R enters through $\log R$, which defeats the precision $o(1/\log n)$.
- (4) Non-equilibrium initial data, such as Wigner matrices run through the matrix Ornstein-Uhlenbeck flow. Since the cover time exceeds the relaxation time, the same limits are expected; hits occurring before relaxation must be controlled.

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